# Practice workbook · Describe a control or explain why a step is in a method

Control tubes, discs or beakers without the active ingredient, and justified processing steps such as blotting, rinsing or using percentage change.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-EXPLAIN-CONTROL-OR-METHOD-STEP

# 1. Question 8 · part b.iii · 3 marks

June 2023 Paper 1R

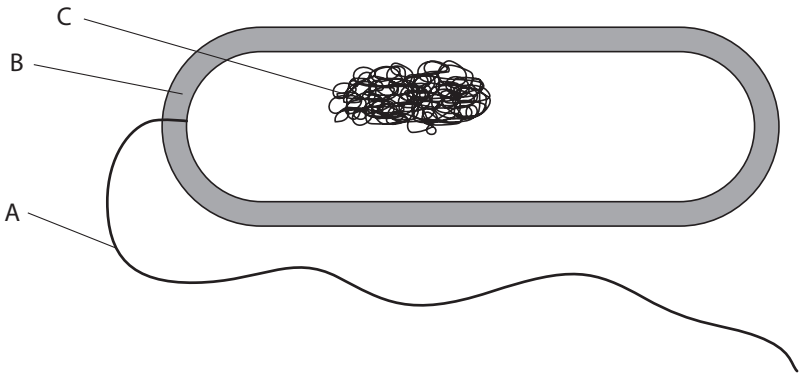
Worked-example retry · Attempt part b.iii only.

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**8** The diagram shows a bacterial cell.




(a) (i) Name the parts labelled A, B and C.

**(3)**

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

**(1)**

.....

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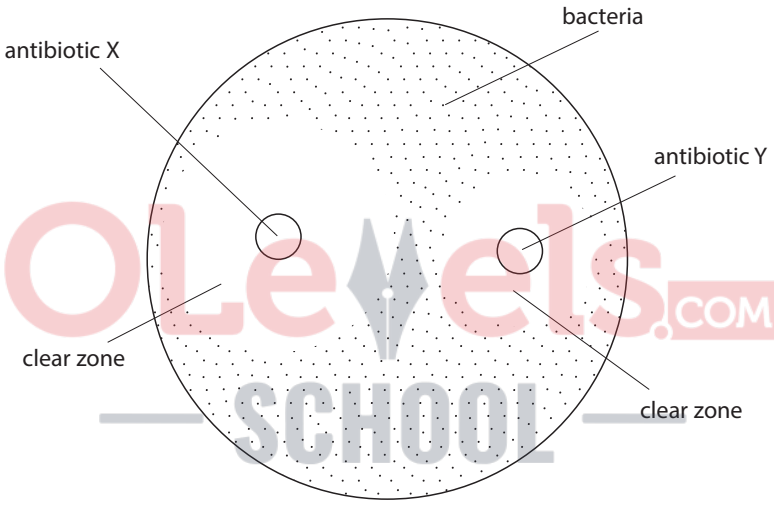
# 1. Question 8 · part b.iii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.iii only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days




(i) Describe what is meant by the term **antibiotic**.

**(2)**

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P 7 2 5 8 6 A 0 1 8 2 0

# 1. Question 8 · part b.iii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.iii only.

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(ii) Explain which antibiotic, X or Y, is more effective against the bacteria in the Petri dish.

**(3)**

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(iii) Explain a suitable control that could be used in this investigation.

**(3)**

**(Total for Question 8 = 12 marks)**

## 1. Question 8 · part b.iii · 3 marks · Working and answer

Working and answer

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## 2. Question 9 · part b.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.ii only.

**9** Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

**(3)**

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

**(3)**

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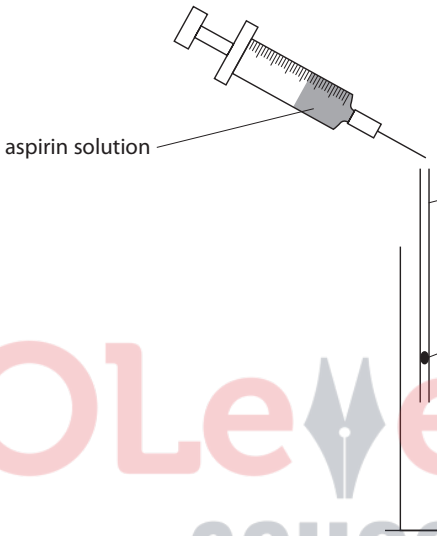
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## 2. Question 9 · part b.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.ii only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.




The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

**(1)**

(ii) Describe a suitable control test for this investigation.

**(1)**

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P 5 8 5 6 6 A 0 2 4 2 8

## 2. Question 9 · part b.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.ii only.

The table shows the student's results.

| Time in minutes | Total volume of aspirin solution in the measuring cylinder in $cm^3$ | | |
|---|---|---|---|
| | 1 aspirin tablet | 2 aspirin tablets | 3 aspirin tablets |
| 5 | 0 | 0 | 0 |
| 10 | 0 | 0 | 0 |
| 15 | 0 | 0 | 1 |
| 20 | 0 | 1 | 2 |
| 25 | 1 | 1 | 3 |
| 30 | 1 | 2 | 4 |
| 35 | 2 | 4 | 6 |
| 40 | 5 | 7 | 9 |
| 45 | 10 | 12 | 16 |
| 50 | 17 | 19 | 22 |
| 55 | 25 | 27 | 31 |
| 60 | 35 | 37 | 39 |

(iii) Explain the pattern shown by the results.

**(3)**

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## 2. Question 9 · part b.ii · 1 mark · Working and answer

Working and answer

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# 3. Question 2 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

**2** A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

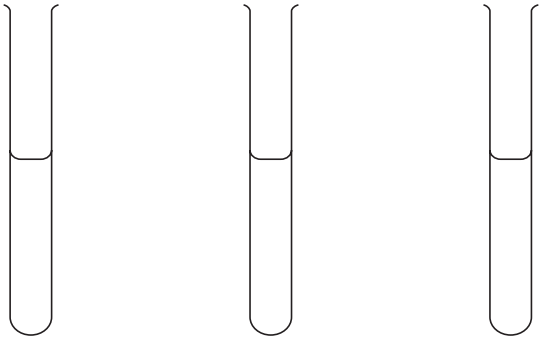



Table 1 lists the contents of each test tube.

| Test tube A | Test tube B | Test tube C |
|---|---|---|
| 5 $cm^3$ of milk | 5 $cm^3$ of milk | 5 $cm^3$ of milk |
| 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate |
| 6 drops of phenolphthalein | 6 drops of phenolphthalein | 6 drops of phenolphthalein |
| 1 $cm^3$ enzyme | 1 $cm^3$ enzyme | 1 $cm^3$ boiled enzyme |
| distilled water | bile salts | bile salts |

**Table 1**

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## 3. Question 2 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

| | Colour of contents | | |
|---|---|---|---|
| | **Tube A** | **Tube B** | **Tube C** |
| at start | pink | pink | pink |
| after 5 minutes | pink | colourless | pink |
| after 10 minutes | pink | colourless | pink |
| after 15 minutes | colourless | colourless | pink |

**Table 2**

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation? **(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids? **(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced? **(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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## 3. Question 2 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

(b) Explain the purpose of tube C.

**(2)**

(c) Suggest one reason for the addition of the sodium hydrogencarbonate.

**(2)**

(d) Describe the effect of bile salts on lipid digestion in this investigation.

**(3)**

**(Total for Question 2 = 10 marks)**

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## 3. Question 2 · part b · 2 marks · Working and answer

Working and answer

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# 4. Question 1 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

**1** The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 $cm^3$ of water, and mix into a smooth paste
- put 10 $cm^3$ of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

*room temperature = 5  heated to 70 °C = 0*

*kept at 3 °C = 1  acid added = 2  alkali added = 3*

*no hydrogen peroxide added = 0*

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# 4. Question 1 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

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(a) (i) Give the student's results in the form of a suitable table.

**(3)**

(ii) State why this investigation is not reliable.

**(1)**

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(iii) Give a reason why the results of this investigation are not accurate.

**(1)**

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

**(3)**

# 4. Question 1 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

(b) (i) State the condition where the enzyme was most active.

**(1)**

(ii) State the reason for including the beaker with no hydrogen peroxide added.

**(1)**

(iii) Explain the result obtained when the conditions were acidic.

**(4)**

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**(Total for Question 1 = 14 marks)**

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Working and answer

## 5. Question 2 · part a.i · 2 marks

June 2022 Paper 2R

Further practice · Attempt part a.i only.

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**2** A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 $cm^3$ of each solution into a different test tube
- add 1.0 $cm^3$ of a competitive inhibitor and 1.0 $cm^3$ of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 $cm^3$ of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

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2 .....

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3 .....

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## 5. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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# 6. Question 6 · part d · 1 mark

June 2022 Paper 2

Further practice · Attempt part d only.

**6** A student uses this method to investigate the effect of different antibacterial cleaners on the growth of bacteria.

- add bacterial culture to sterile agar in a Petri dish using an inoculating loop
- using a pipette, place two drops of an antibacterial cleaner onto a filter paper disc
- repeat using two other filter paper discs, each one soaked with a different antibacterial cleaner
- soak one filter paper disc in distilled water
- place the four filter paper discs onto the sterile agar
- place a lid on the Petri dish
- place the dish into an incubator at 25 °C for 5 days

(a) State the independent variable in this investigation.

**(1)**

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(b) Give two improvements to this method that will reduce the risk of introducing unwanted microorganisms into the Petri dish.

**(2)**

1 .....

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2 .....

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## 6. Question 6 · part d · 1 mark

June 2022 Paper 2

Further practice · Attempt part d only.

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(c) Explain why it is important that the Petri dish is incubated at a temperature no higher than 25 °C.

**(2)**

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(d) Give a reason why the filter paper disc soaked in distilled water is included in the investigation.

**(1)**

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**(Total for Question 6 = 6 marks)**

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Working and answer

# 7. Question 7 · part d · 3 marks

November 2024 Paper 2

Further practice · Attempt part d only.

**7** Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

| Student | Volume of enzyme solution in $cm^3$ | Temperature in °C | pH | Volume of cloudy suspension in $cm^3$ | Time tubes left in minutes | Appearance at end of investigation |
|---|---|---|---|---|---|---|
| 1 | 5 | 37 | 2 | 10 | 9 | cloudy |
| 2 | 5 | 37 | 2 | 10 | 20 | clear |
| 3 | 5 | 37 | 4 | 10 | 35 | clear |
| 4 | 5 | 37 | 4 | 10 | 20 | cloudy |
| 5 | 5 | 37 | 6 | 10 | 8 | cloudy |
| 6 | 5 | 37 | 6 | 10 | 40 | cloudy |

(a) Explain why the investigation was carried out at 37 °C.

**(2)**

(b) Explain which property of the enzyme the students are investigating.

**(2)**

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# 7. Question 7 · part d · 3 marks

November 2024 Paper 2

Further practice · Attempt part d only.

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(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

**(4)**

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(d) Describe a control experiment that the students could have carried out.

**(3)**

# 7. Question 7 · part d · 3 marks

November 2024 Paper 2

Further practice · Attempt part d only.

(e) State a possible conclusion that the students could make from the results of this investigation.

**(2)**

(f) Explain where this enzyme is likely to be found in the alimentary canal.

**(2)**

**(Total for Question 7 = 15 marks)**

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## 7. Question 7 · part d · 3 marks · Working and answer

Working and answer

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# 8. Question 5 · part c.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

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**5** Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

**(2)**

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

**(5)**

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# 8. Question 5 · part c.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

(c) The table shows results obtained from an investigation to determine the concentration of the contents of potato cells.

| | Concentration of sucrose solution in mol per dm³ | | | | |
|---|---|---|---|---|---|
| | **0.2** | **0.4** | **0.6** | **0.8** | **1.0** |
| Initial mass in g | 8.4 | 7.9 | 8.7 | 8.3 | 8.4 |
| Mass after soaking in g | 8.7 | 8.0 | 8.2 | 7.5 | 7.3 |
| Change in mass in g | + 0.3 | + 0.1 | – 0.5 | – 0.8 | – 1.1 |
| Percentage change in mass | + 3.6 | | – 5.8 | – 9.6 | – 13.1 |

(i) Calculate the percentage change in mass for 0.4 mol per dm³.

**(2)**

.....%

(ii) Explain why it is necessary to calculate the percentage change in mass rather than using the change in mass in g.

**(2)**

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# 8. Question 5 · part c.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

(iii) Explain how the results in the table could be used to determine the concentration of the contents of potato cells.

**(4)**

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**(Total for Question 5 = 15 marks)**

Working and answer

# 9. Question 4 · part c.i · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.i only.

**4** A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

| | | |
|---|---|---|
| 1.00 au = −7 % | 1.50 au = −10 % | 0.00 au (distilled water) = +8 % |
| | 0.25 au = +4 % | 0.50 au = −1 % |

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

**(4)**

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# 9. Question 4 · part c.i · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.i only.

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(ii) Determine the median value for sugar concentration.

**(1)**

median value = .....

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

**(3)**

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mass of cylinder = ..... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

**(4)**

concentration of sugar solution = ..... au

# 9. Question 4 · part c.i · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.i only.

(c) (i) Explain why filter paper was wrapped around the potato cylinders at the beginning and end of the investigation.

**(2)**

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(ii) The student kept the potato cylinders in the liquid for two hours.

State two other factors that should be kept constant during this investigation.

**(2)**

1 .....

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2 .....

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**(Total for Question 4 = 16 marks)**

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Working and answer

# 10. Question 2 · part a.iii · 3 marks

January 2022 Paper 1

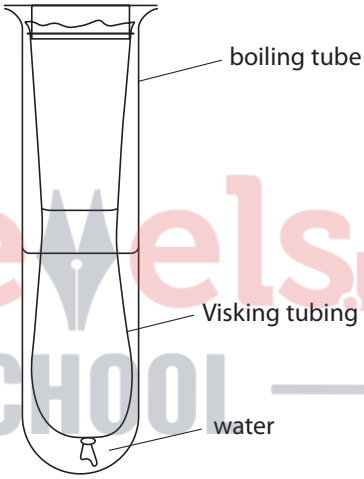
Further practice · Attempt part a.iii only.

**2** A student investigates absorption in the ileum by using a model gut made of Visking tubing.

This is the student's method.

- tie a knot in one end of a 15 cm length of Visking tubing to make a bag
- place 5 $cm^3$ of starch suspension and 5 $cm^3$ of glucose solution into the bag
- rinse the outside of the bag with water
- place the bag into a boiling tube containing distilled water
- leave the bag for one hour

The diagram shows the student's apparatus.




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After an hour the contents of the Visking tubing bag are tested for glucose and starch.

The water in the boiling tube is also tested for glucose and starch.

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## 10. Question 2 · part a.iii · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

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(a) (i) Describe how the student should test the contents of the Visking tubing bag for glucose.

**(4)**

(ii) Describe how the student should test the contents of the Visking tubing bag for starch.

**(3)**

(iii) Explain why the outside of the Visking tubing bag was rinsed with water before being placed in the boiling tube.

**(3)**

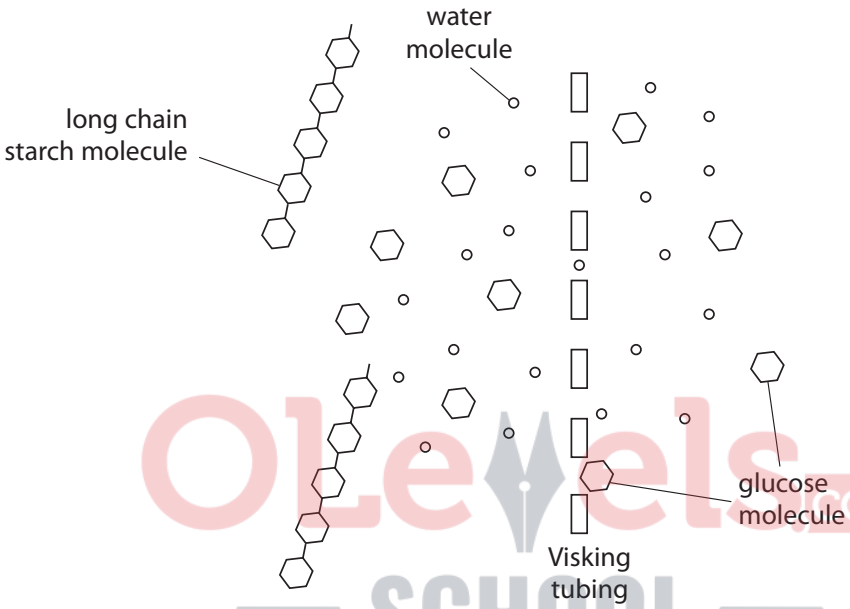
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## 10. Question 2 · part a.iii · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

(b) The student draws this diagram to explain the results obtained.




Explain the results the student obtains.

Refer to the diagram in your answer.

**(5)**

**(Total for Question 2 = 15 marks)**

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## 10. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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